ENVS 193DS — Homework 3

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Set Up

```r
# reading in relevant packages
library(tidyverse)
library(janitor)
library(here)
library(readxl)
library(patchwork)

# storing kelp/temp data in object 'kelp'
kelp <- read.csv(here("data","temp-kelp.csv"))
# storing personal data in object 'exercise'
exercise <- read.csv(here("data","exercise.csv"))
# setting global theme for plots as theme_bw
theme_set(theme_bw())
```

Problems

Problem 1. Giant Kelp Fronds

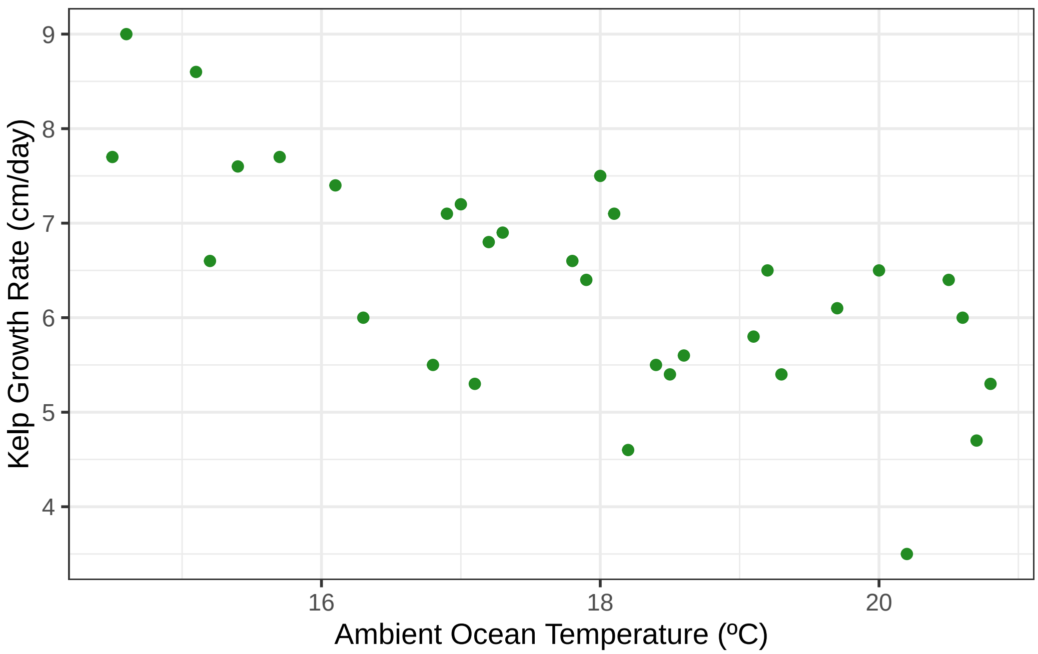
a. Choosing an Appropriate Test

The appropriate test to determine the strength of the relationship between temperature and giant kelp frond elongation rate is a Pearson Correlation test. This test is appropriate because both variables of interest represent continuous, numerical data that fit into this parametric test.

b. Creating a Visualization

```r
# base layer ggplot
ggplot(
  # using kelp data
  data = kelp,
  # setting axes
  mapping = aes(x = temp_c,
                y = kelp_elong)) +
  # representing individual observations as points
  geom_point(color = 'forestgreen') +
```

```
  # relabelling axes
  labs(x = "Ambient Ocean Temperature (ºC)",
       y = "Kelp Growth Rate (cm/day)")
```



c. Checking Assumptions and Running Test

```
# creating temperature qq plot using base layer ggplot
qq_temp <- ggplot(
  # using kelp data
  data = kelp,
  # setting sample to temperature
  mapping = aes(sample = temp_c)) +
  # adding QQ line
  geom_qq_line() +
  # adding qq points and coloring blue
  geom_qq(col = 'blue') +
  # removing axis labels and adding title
  labs(x = "",
       y = "",
       title = "Ocean Temperature")
```

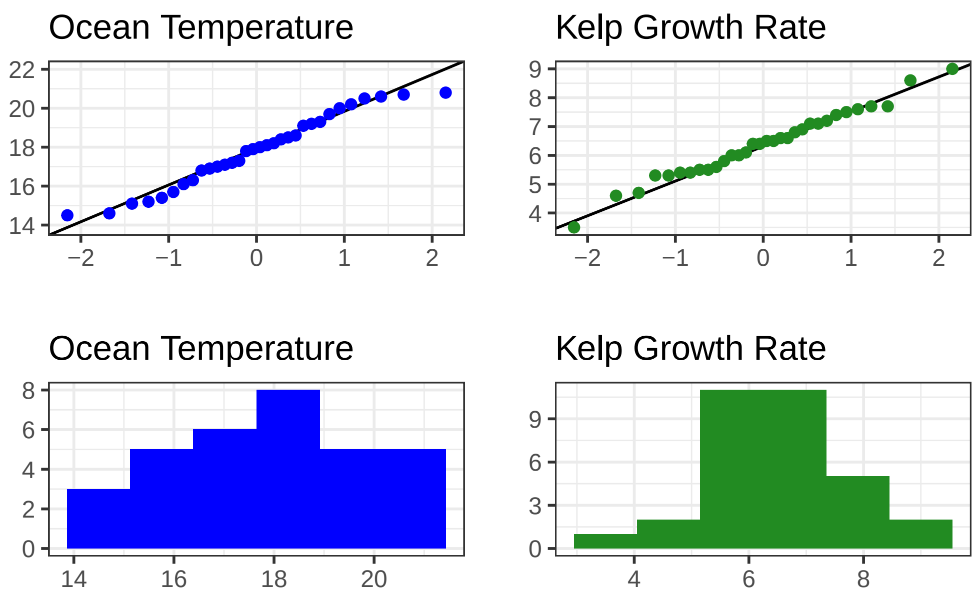
```r
# creating kelp qq plot using base layer ggplot
qq_kelp <- ggplot(
  # using kelp data
  data = kelp,
  # setting sample to temperature
  mapping = aes(sample = kelp_elong)) +
  # adding QQ line
  geom_qq_line() +
  # adding qq points and coloring blue
  geom_qq(col = 'forestgreen') +
  # removing axis labels and adding title
  labs(x = "",
       y = "",
       title = "Kelp Growth Rate")

# creating temperature histogram using base layer ggplot
histogram_temp <- ggplot(
  # using data from object 'kelp'
  data = kelp,
  # setting axis as temperature
  mapping = aes(x = temp_c)) +
  # creating blue histogram
  geom_histogram(fill = 'blue',
                 # using rice rule for bin number
                 bins = 6) +
  # removing axis labels and adding title
  labs(x = "",
       y = "",
       title = "Ocean Temperature")

# creating kelp histogram using base layer ggplot
histogram_kelp <- ggplot(
  # using data from object 'kelp'
  data = kelp,
  # setting axis as kelp elongation rate
  mapping = aes(x = kelp_elong)) +
  # creating green histogram
  geom_histogram(fill = 'forestgreen',
                 # using rice rule for bin number
                 bins = 6) +
  # removing axis labels and adding title
  labs(x = "",
```

```
       y = "",
       title = "Kelp Growth Rate")

# displaying above plots in 2x2 grid using patchwork package
(qq_temp + qq_kelp)/ (histogram_temp + (histogram_kelp))
```




```
# displaying 10 random rows from kelp data
slice_sample(kelp, n = 10)
```

```
   temp_c kelp_elong
1    15.7        7.7
2    15.2        6.6
3    19.2        6.5
4    18.4        5.5
5    20.8        5.3
6    15.4        7.6
7    18.0        7.5
8    16.1        7.4
9    18.5        5.4
10   18.1        7.1
```

```
# performing a correlation test between kelp growth rate and ocean temperature
cor.test(
  # formula: variable, variable
  kelp$kelp_elong, kelp$temp_c,
         # specifying pearson's correlation
         method = "pearson")
```

```

    Pearson's product-moment correlation

data:  kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

I checked the following assumptions necessary for a pearson correlation:

1. Continuity
2. Normal distribution
3. Linear Relationship
4. Independent Observations

I checked these assumptions using the following methods:

1. I displayed 10 random rows and checked for decimal measurements. I also thought critically about what kind of data I'm using; Both temperature and growth rates are infinitely divisible.
2. I visually assessed the normality of the distribution of both variables using QQ plots and histograms.
3. I visually assessed linearity between both variables using a scatterplot.
4. I read through the metadata where it explicitly states "individuals".

Through these methods I assessed the data to be both continuous and relatively normally distributed. While the temperature data are heavy-tailed, it is still unimodal and the majority of QQ points fall along the reference line. A negative, linear relationship was also relatively clear using a scatterplot. I am trusting that the observations are independent of eachother since I am told each individual is tracked, and no repeated observations are mentioned.

d. Communicating Results

To evaluate the strength of the relationship between ocean temperature and giant kelp frond growth rate, I utilized a pearson's product-moment correlation test.

We found a moderate, negative relationship between ocean temperature and giant kelp frond growth rate (Pearson's r = -0.688, t(30) = -5.2, $p < 0.001$, $\alpha = 0.05$). Therefore, we reject the null hypothesis that there is no relationship between ocean temperature and kelp growth rates.

e. Test Implications

Through a Pearson's correlation test, I found there to be a moderately strong correlation between ocean temperature and giant kelp frond growth rates. Growth rates diplayed a negative linear relationship with ocean temperature, with kelp thriving in temperatures below 16ºC. The lower boundary of my test was 14ºC, therefore further research in colder environments may be warranted.

###f. Double Checking Work

```r
# performing a correlation test between kelp growth rate and ocean temperature
cor.test(
  # formula: variable, variable
  kelp$temp_c, kelp$kelp_elong,
  # specifying spearman's rank correlation test
         method = "spearman")
```

```
	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho 
-0.6891684 
```

Both Spearman and Pearson correlation tests would have led me to reject the null hypothesis since both tests have p-values much less than my significance level of 0.05. Both tests also describe the relationship between ocean temperature and giant kelp frond growth rate as negative, linear and moderately influential (Pearson's r = -0.688, Spearman's ρ = -0.689).

Problem 2. Personal Data

a. Updating Visualizations

b. Creating Captions

Problem 3. Affective Visualization

a. Describing Affective Visualization for Personal Data

b. Creating Paper Sketch of Idea

c. Drafting a Visualization

d. Writing an Artist Statement

e. Prepping materials to share in class

Problem 4. Statistical Critique

a. Revisiting and Summarizing

b. Visual Clarity

c. Aesthetic Clarity

d. Recommendations